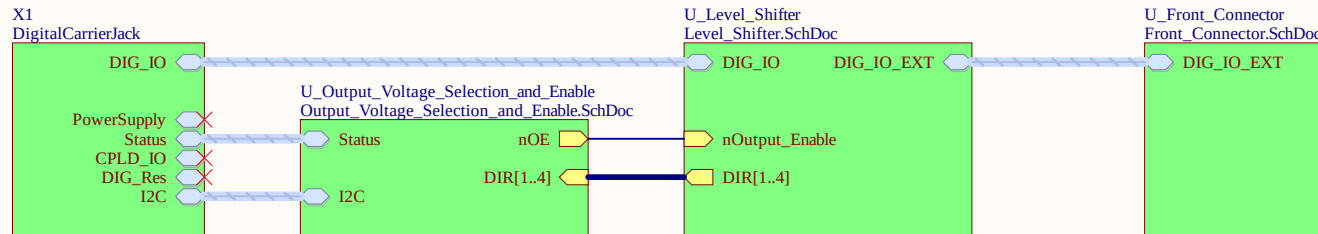


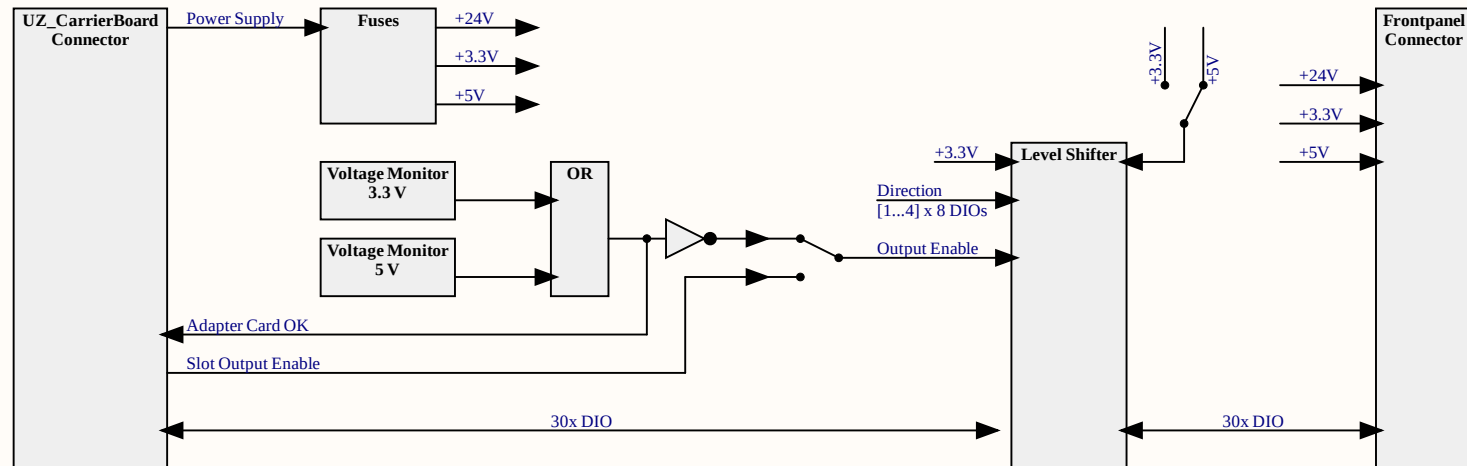
Top Sheet



Block Diagram

Board Features:

- configurable output voltage level (3.3V or 5V)
- voltage monitor ICs supervise 3.3V and 5V supply rail
- output enable of level shifters can be controlled either by the supervisors status or via UZ_CarrierBoard signal
- I²C-Port Expander reads selected output voltage level and signal direction per group of 8 channels
- RGB-LED at front panel visualizes selected voltage and error status



Chose size of UZ logo
by choosing footprint
type



LOG01

INFO

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

Title TopSheet.SchDoc

Revision: Rev02

Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PrjPCB

UltraZohm

www.ultrazohm.com

Date: 01.04.2025

Sheet 1 of 5

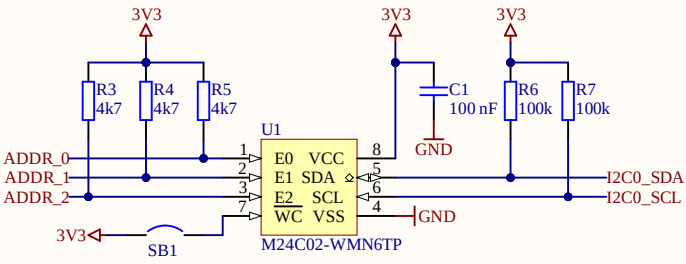
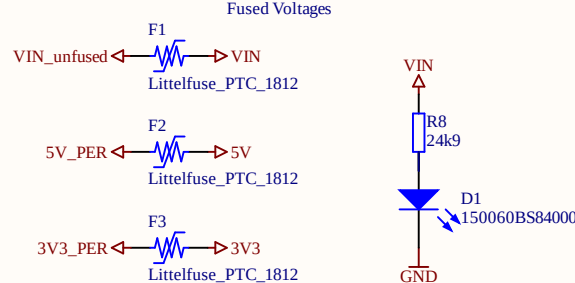
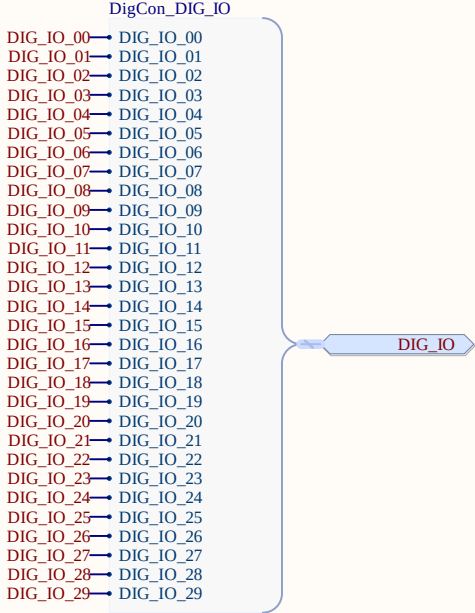
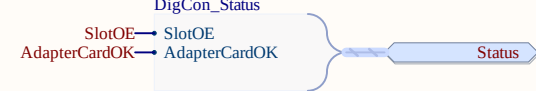
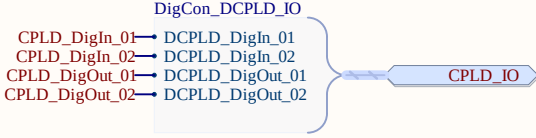
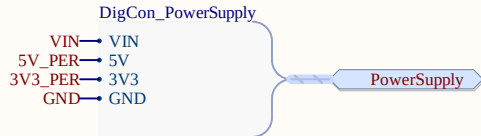
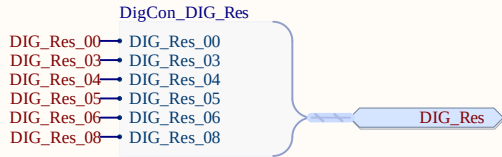
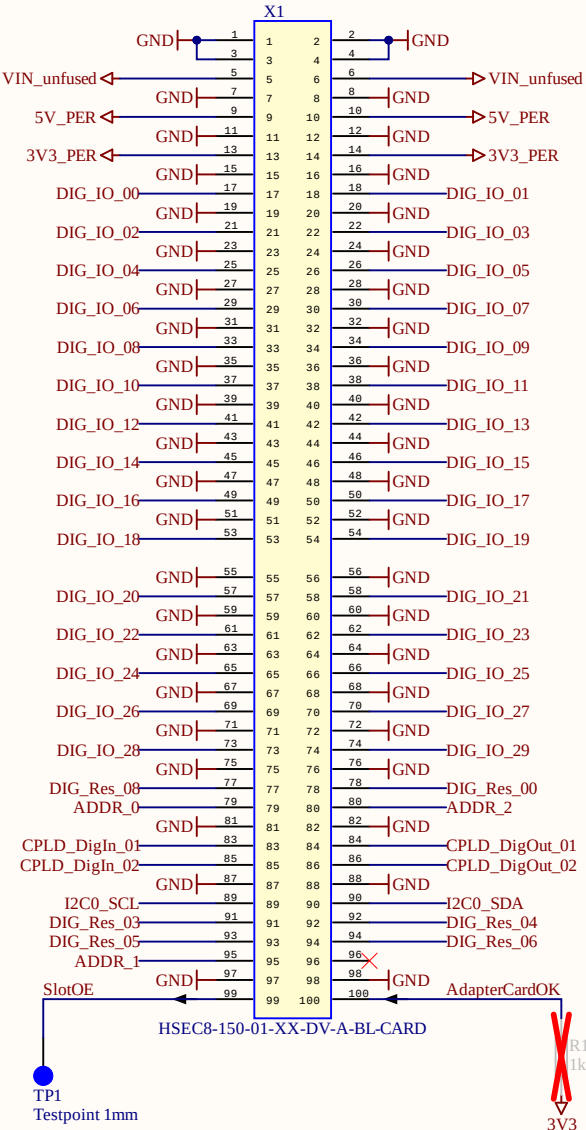


A

B

C

D



Note: place solder bump to protect EEPROM memory from write access.

PC-Address: 0b101 E2 E1 E0 = 0h5[0-7]

E2/E1/E0 is configured on UZ_CarrierBoard hardware related to the used slot D[1..5].

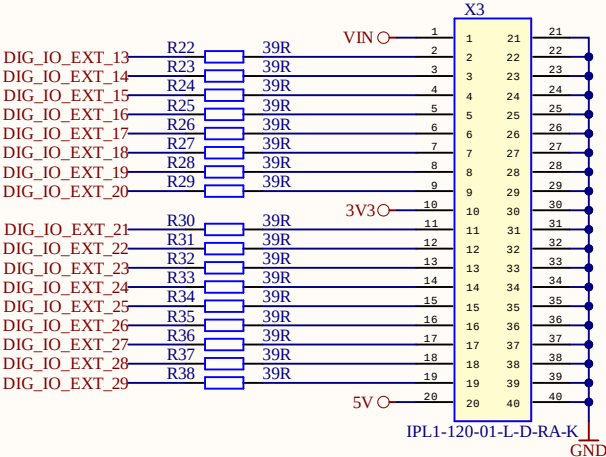
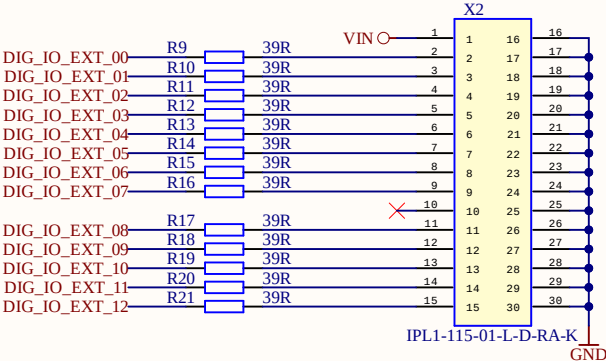
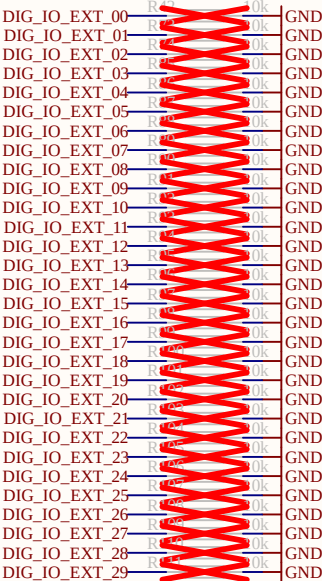
Title DigitalCarrierJack.SchDoc		UltraZohm www.ultrazohm.com	
Revision: Rev02	Design Engineer: A. Geiger		
Project: UZ_D_Voltage_3V3_5V.PrjPCB			
		Date: 01.04.2025	
		Sheet 2 of 5	

Pull-Down Resistors

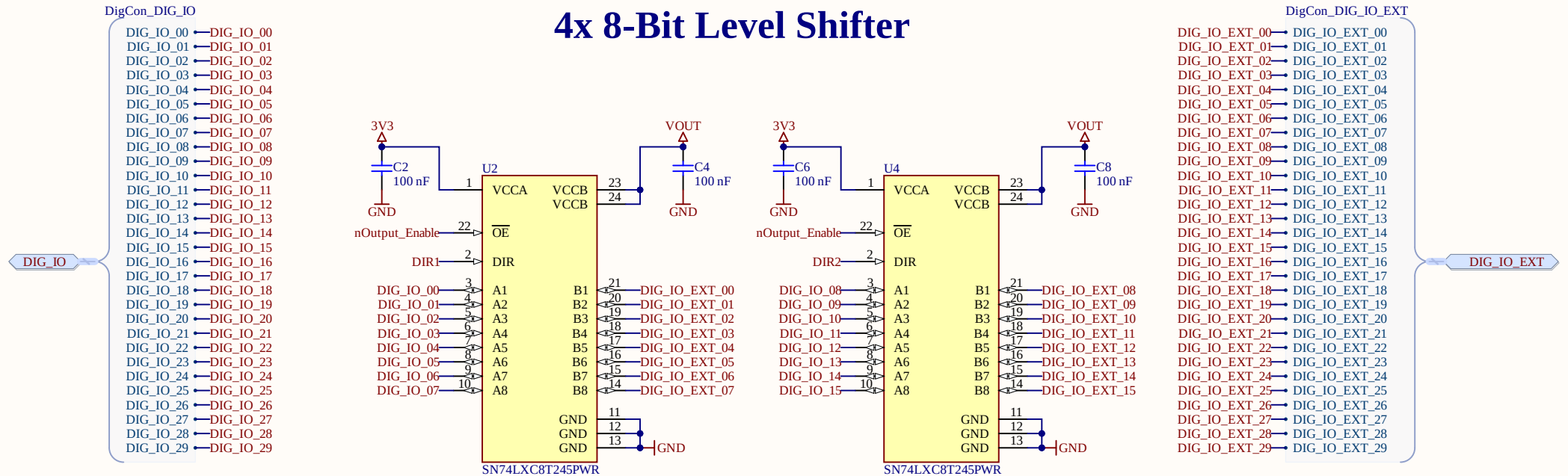
Front Panel Connector

DigCon_DIG_IO_EXT

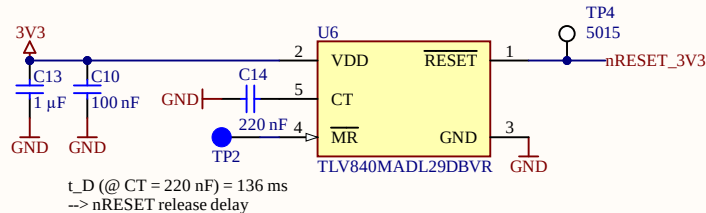
DIG_IO_EXT_00 → DIG_IO_EXT_00
DIG_IO_EXT_01 → DIG_IO_EXT_01
DIG_IO_EXT_02 → DIG_IO_EXT_02
DIG_IO_EXT_03 → DIG_IO_EXT_03
DIG_IO_EXT_04 → DIG_IO_EXT_04
DIG_IO_EXT_05 → DIG_IO_EXT_05
DIG_IO_EXT_06 → DIG_IO_EXT_06
DIG_IO_EXT_07 → DIG_IO_EXT_07
DIG_IO_EXT_08 → DIG_IO_EXT_08
DIG_IO_EXT_09 → DIG_IO_EXT_09
DIG_IO_EXT_10 → DIG_IO_EXT_10
DIG_IO_EXT_11 → DIG_IO_EXT_11
DIG_IO_EXT_12 → DIG_IO_EXT_12
DIG_IO_EXT_13 → DIG_IO_EXT_13
DIG_IO_EXT_14 → DIG_IO_EXT_14
DIG_IO_EXT_15 → DIG_IO_EXT_15
DIG_IO_EXT_16 → DIG_IO_EXT_16
DIG_IO_EXT_17 → DIG_IO_EXT_17
DIG_IO_EXT_18 → DIG_IO_EXT_18
DIG_IO_EXT_19 → DIG_IO_EXT_19
DIG_IO_EXT_20 → DIG_IO_EXT_20
DIG_IO_EXT_21 → DIG_IO_EXT_21
DIG_IO_EXT_22 → DIG_IO_EXT_22
DIG_IO_EXT_23 → DIG_IO_EXT_23
DIG_IO_EXT_24 → DIG_IO_EXT_24
DIG_IO_EXT_25 → DIG_IO_EXT_25
DIG_IO_EXT_26 → DIG_IO_EXT_26
DIG_IO_EXT_27 → DIG_IO_EXT_27
DIG_IO_EXT_28 → DIG_IO_EXT_28
DIG_IO_EXT_29 → DIG_IO_EXT_29



4x 8-Bit Level Shifter

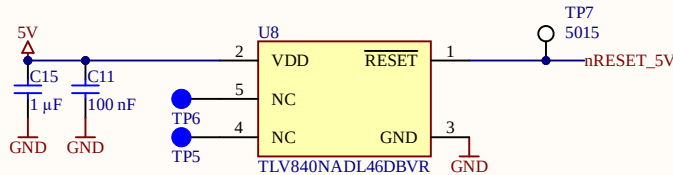


3.3 V Voltage Monitoring



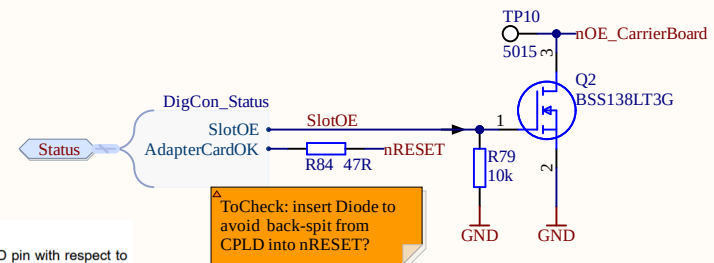
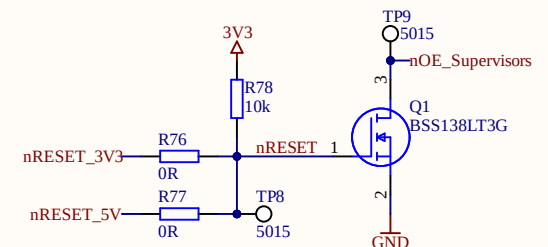
3V3 Rail	/RESET
0 ... 0,7 V	"LOW" (follows 3V3 rail voltage)
0,7 ... 2,9 V	LOW
2,9 ... 3,3 V	HIGH

5 V Voltage Monitoring

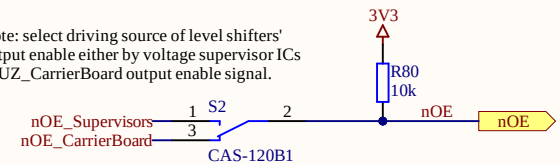


5V Rail	/RESET
0 ... 0,7 V	"LOW" (follows 3V3 rail voltage)
0,7 ... 4,6 V	LOW
4,6 ... 5 V	HIGH

Enable Signal Logic Inversion



Note: select driving source of level shifters' output enable either by voltage supervisor ICs or UZ_CarrierBoard output enable signal.



Voltage Monitoring Functionality

8.4 Device Functional Modes

Table 8-1 summarizes the various functional modes of the device. Logic high is represented by "H" and logic low is represented by "L".

Table 8-1. Truth Table

VDD	MR	RESET	RESET
VDD < V _{POR}	Ignored	Undefined	Undefined
V _{POR} < V _{DD} < V _{IT}	Ignored	H	L
VDD ≥ V _{IT}	L	H	L
VDD ≥ V _{IT}	H	L	H
VDD ≥ V _{IT}	Floating	L	H

8.4.1 Normal Operation (V_{DD} > V_{POR})

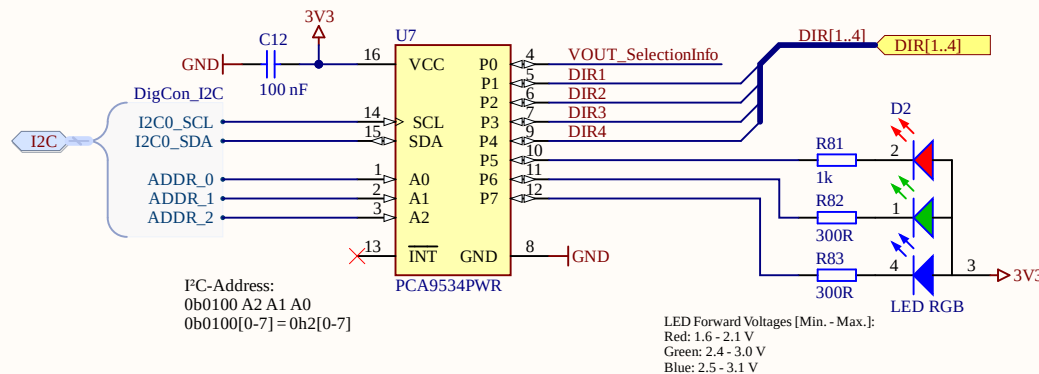
When VDD is greater than VPOR, the reset signal is determined by the voltage on the VDD pin with respect to the trip point (V_{IT}).

- MR high: the reset signal corresponds to VDD with respect to the threshold voltage.
- MR low: in this mode, the reset is asserted regardless of the threshold voltage.

8.4.2 Below Power-On-Reset (V_{DD} < V_{POR})

When the voltage on VDD is lower than VPOR, the device does not have enough bias voltage to internally pull the asserted output low or high and reset voltage level is undefined.

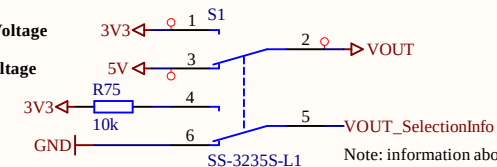
I²C-Port Expander



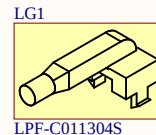
Level Shifter - Output Voltage Selection

use 3.3 V Output Voltage

use 5 V Output Voltage



Note: information about selected output voltage
HIGH --> VOUT = 3.3 V
LOW --> VOUT = 5 V



Title Output_Voltage_Selection_and_Enable.SchDoc

Revision: Rev02 Design Engineer: A. Geiger

Project: UZ_D_Voltage_3V3_5V.PrjPCB

UltraZohm

www.ultrazohm.com

Date: 01.04.2025

Sheet 5 of 5